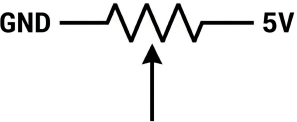


Resistors

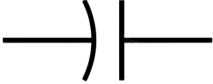
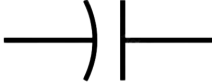
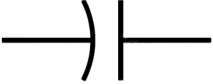
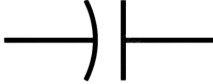
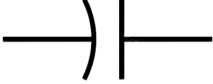
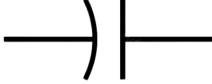
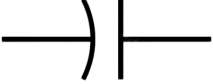
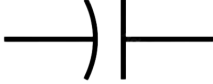
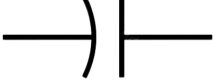
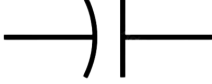
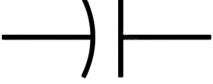
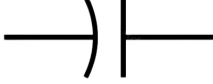
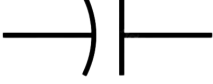
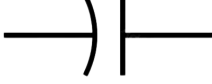
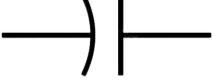
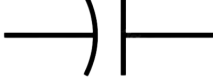
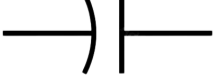
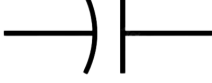
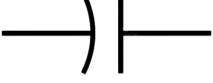
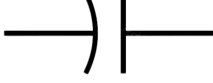
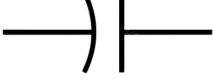
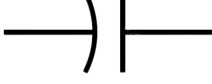
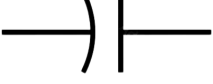
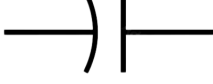
10Ω  Through-Hole Resistor ±1% tolerance	100Ω  Through-Hole Resistor ±1% tolerance	220Ω  Through-Hole Resistor ±1% tolerance	330Ω  Through-Hole Resistor ±1% tolerance
1kΩ  Through-Hole Resistor ±1% tolerance	2kΩ  Through-Hole Resistor ±1% tolerance	5.1kΩ  Through-Hole Resistor ±1% tolerance	10kΩ  Through-Hole Resistor ±1% tolerance
100kΩ  Through-Hole Resistor ±1% tolerance	1MΩ  Through-Hole Resistor ±1% tolerance	10Ω  Precision Resistor ±0.1% tolerance	100Ω  Precision Resistor ±0.1% tolerance
1kΩ  Precision Resistor ±0.1% tolerance	Potentiometer  Variable Resistor 1: GND 2: Vout 3: 5V	500Ω  Trimmer Potentiometer RM063 Variable Resistor Code: 501	1kΩ  Trimmer Potentiometer RM063 Variable Resistor Code: 102
2kΩ  Trimmer Potentiometer RM063 Variable Resistor Code: 202	5kΩ  Trimmer Potentiometer RM063 Variable Resistor Code: 502	10kΩ  Trimmer Potentiometer RM063 Variable Resistor Code: 103	20kΩ  Trimmer Potentiometer RM063 Variable Resistor Code: 203

50kΩ  Trimmer Potentiometer RM063 Variable Resistor Code: 503	100kΩ  Trimmer Potentiometer RM063 Variable Resistor Code: 104	200kΩ  Trimmer Potentiometer RM063 Variable Resistor Code: 204	1MΩ  Trimmer Potentiometer RM063 Variable Resistor Code: 105
---	---	--	---

Ceramic Capacitors

10pF  Ceramic Capacitor Non-polarized	20pF  Ceramic Capacitor Non-polarized	30pF  Ceramic Capacitor Non-polarized	47pF  Ceramic Capacitor Non-polarized
56pF  Ceramic Capacitor Non-polarized	68pF  Ceramic Capacitor Non-polarized	100pF  Ceramic Capacitor Non-polarized	220pF  Ceramic Capacitor Non-polarized
330pF  Ceramic Capacitor Non-polarized	680pF  Ceramic Capacitor Non-polarized	1nF  Ceramic Capacitor Non-polarized	4.7nF  Ceramic Capacitor Non-polarized
10nF  Ceramic Capacitor Non-polarized	47nF  Ceramic Capacitor Non-polarized	100nF  Ceramic Capacitor Non-polarized Decoupling / filtering	

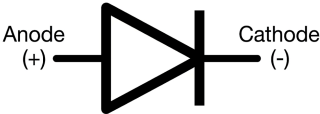
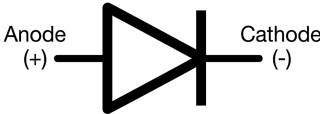
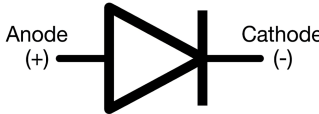
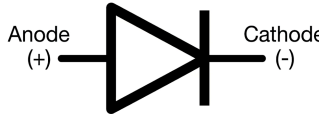
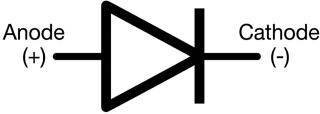
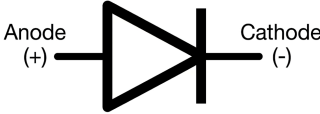
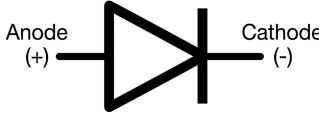
Electrolytic Capacitors

0.1μF 50V  Electrolytic Capacitor Polarized Long lead: +	0.22μF 50V  Electrolytic Capacitor Polarized Long lead: +	0.47μF 50V  Electrolytic Capacitor Polarized Long lead: +	1μF 50V  Electrolytic Capacitor Polarized Long lead: +
2.2μF 50V  Electrolytic Capacitor Polarized Long lead: +	3.3μF 50V  Electrolytic Capacitor Polarized Long lead: +	4.7μF 50V  Electrolytic Capacitor Polarized Long lead: +	10μF 25V  Electrolytic Capacitor Polarized Long lead: +
10μF 50V  Electrolytic Capacitor Polarized Long lead: +	22μF 50V  Electrolytic Capacitor Polarized Long lead: +	22μF 25V  Electrolytic Capacitor Polarized Long lead: +	33μF 25V  Electrolytic Capacitor Polarized Long lead: +
47μF 10V  Electrolytic Capacitor Polarized Long lead: +	47μF 25V  Electrolytic Capacitor Polarized Long lead: +	47μF 50V  Electrolytic Capacitor Polarized Long lead: +	100μF 16V  Electrolytic Capacitor Polarized Long lead: +
100μF 25V  Electrolytic Capacitor Polarized Long lead: +	220μF 10V  Electrolytic Capacitor Polarized Long lead: +	220μF 25V  Electrolytic Capacitor Polarized Long lead: +	330μF 25V  Electrolytic Capacitor Polarized Long lead: +
470μF 10V  Electrolytic Capacitor Polarized Long lead: +	470μF 16V  Electrolytic Capacitor Polarized Long lead: +	680μF 16V  Electrolytic Capacitor Polarized Long lead: +	1000μF 16V  Electrolytic Capacitor Polarized Long lead: +

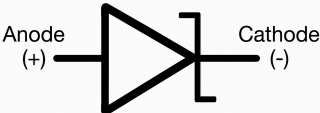
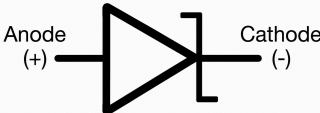
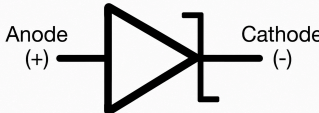
Air Core Inductors

0.7×3.5×1.5T  Air-Core RF Inductor Measure μH if needed	0.7×3.5×2.5T  Air-Core RF Inductor Measure μH if needed	0.7×3.5×3.5T  Air-Core RF Inductor Measure μH if needed	0.7×3.5×4.5T  Air-Core RF Inductor Measure μH if needed
0.7×3.5×5.5T  Air-Core RF Inductor Measure μH if needed	0.7×3.5×6.5T  Air-Core RF Inductor Measure μH if needed	0.7×3.5×7.5T  Air-Core RF Inductor Measure μH if needed	

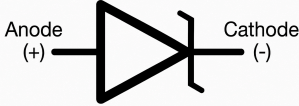
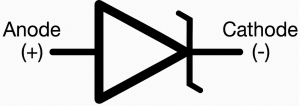
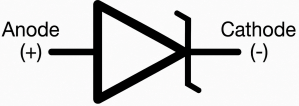
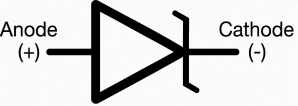
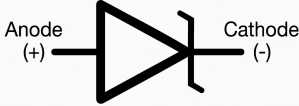
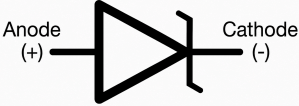
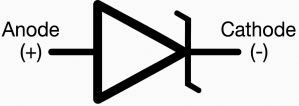
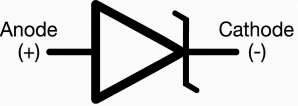
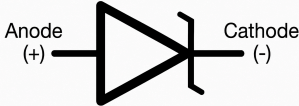
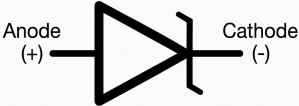
Rectifier Diodes

1N4001  Rectifier Diode Cathode = stripe side VRRM: 50V IF: 1A	1N4002  Rectifier Diode Cathode = stripe side VRRM: 100V IF: 1A	1N4003  Rectifier Diode Cathode = stripe side VRRM: 200V IF: 1A	1N4004  Rectifier Diode Cathode = stripe side VRRM: 400V IF: 1A
1N4005  Rectifier Diode Cathode = stripe side VRRM: 600V IF: 1A	1N4006  Rectifier Diode Cathode = stripe side VRRM: 800V IF: 1A	1N4007  Rectifier Diode Cathode = stripe side VRRM: 1000V IF: 1A	

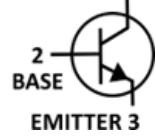
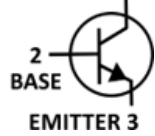
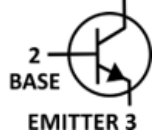
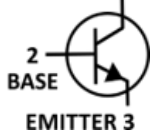
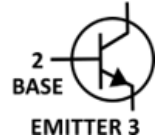
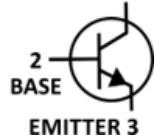
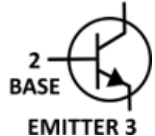
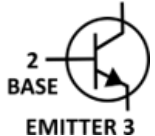
Schottky Diodes

1N5817  Schottky Diode Cathode = stripe side VRRM: 20V IF: 1A	1N5818  Schottky Diode Cathode = stripe side VRRM: 30V IF: 1A	1N5819  Schottky Diode Cathode = stripe side VRRM: 40V IF: 1A	
---	--	---	--

Zener Diodes

1N4728 3.3V  Zener Diode Cathode = stripe side P: 1W	1N4729 3.6V  Zener Diode Cathode = stripe side P: 1W	1N4730 3.9V  Zener Diode Cathode = stripe side P: 1W	1N4731 4.3V  Zener Diode Cathode = stripe side P: 1W
1N4732 4.7V  Zener Diode Cathode = stripe side P: 1W	1N4733 5.1V  Zener Diode Cathode = stripe side P: 1W	1N4734 5.6V  Zener Diode Cathode = stripe side P: 1W	1N4735 6.2V  Zener Diode Cathode = stripe side P: 1W
1N4736 6.8V  Zener Diode Cathode = stripe side P: 1W	1N4737 7.5V  Zener Diode Cathode = stripe side P: 1W		

Bipolar Junction Transistors (NPN)

2N2222  NPN BJT Transistor General switching	2N3904  NPN BJT Transistor General switching/amplifier	2N5551  NPN BJT Transistor High-voltage amplifier	BC337  NPN BJT Transistor General amplifier/switch
BC547  NPN BJT Transistor Small-signal amplifier	BC550  NPN BJT Transistor Low-noise amplifier	C945  NPN BJT Transistor General amplifier	C1815  NPN BJT Transistor Audio/general amplifier

S8050 NPN BJT Transistor General switching	S9013 NPN BJT Transistor General amplifier/switch	S9014 NPN BJT Transistor Small-signal amplifier	S9018 NPN BJT Transistor RF/small-signal amplifier
--	---	---	--

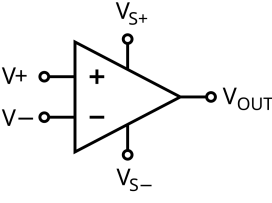
Bipolar Junction Transistors (PNP)

2N2907 PNP BJT Transistor General switching	2N3906 PNP BJT Transistor General switching/amplifier	2N5401 PNP BJT Transistor High-voltage amplifier	BC327 PNP BJT Transistor General amplifier/switch
BC557 PNP BJT Transistor Small-signal amplifier	BC560 PNP BJT Transistor Low-noise amplifier	A1015 PNP BJT Transistor Small-signal amplifier	A733 PNP BJT Transistor General amplifier
S8550 PNP BJT Transistor General switching	S9012 PNP BJT Transistor General amplifier/switch	S9015 PNP BJT Transistor Small-signal amplifier	

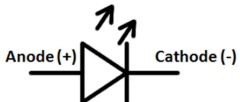
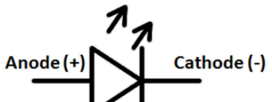
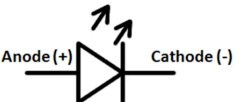
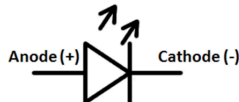
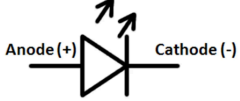
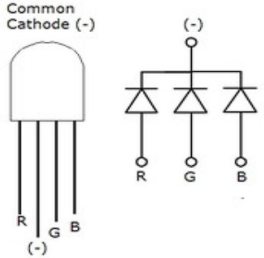
Metal Oxide Semiconductor Field-Effect Transistors

N-Channel MOSFETs Field Effect Transistors Contains IRF530N IRF540N IRF640N IRF740 IRF840	N-Channel MOSFETs Field Effect Transistors Contains 2N7000 IRFZ44N IRF3205 IRF510N IRF520N	P-Channel MOSFETs Field Effect Transistors Contains IRF9540N
---	--	--

Integrated Circuits

Op-Amps / Comparators Analog Signal ICs  Contains - LM358 - LM324 - MCP6002 - NE5532 - JRC4558 - LM393 - LM339	Logic / Timer ICs Digital Utility ICs Timing and output expansion Contains: NE555, 74HC595	Driver ICs Load / Motor Driver ICs Higher-current output control Contains: L293D, ULN2003, ULN2803 ULN2003 Driver Module
Audio ICs / Modules Audio Circuit ICs Amplification, delay, and playback Contains: LM386, TDA2030AN, TDA2822D, PT2399, DFPlayer Mini	Power / Isolation ICs Power Control / Isolation ICs Charge pump, SMPS control, and signal isolation Contains: ICL7660S, UC3842AN, UC3843AN, PC817, 4N35	ADC / DAC Modules Data Conversion Modules Analog input and analog output Contains: MCP4725, ADS1115

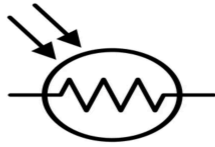
LEDs / Displays

Red LED  Light-Emitting Diode V_f: ~2.0–2.2V I_f max: 20mA	Yellow LED  Light-Emitting Diode V_f: ~2.0–2.2V I_f max: 20mA	Green LED  Light-Emitting Diode V_f: ~2.0–3.2V I_f max: 20mA	Blue LED  Light-Emitting Diode V_f: ~3.0–3.4V I_f max: 20mA
White LED  Light-Emitting Diode V_f: ~3.0–3.2V I_f max: 20mA	RGB LED  Light-Emitting Diode	LED Bar Graph 10-segment LED display Level indicator Use resistors	I2C LCD 1602 16×2 LCD Display I2C interface SDA + SCL
20×4 I2C LCD 20×4 LCD Display I2C interface SDA + SCL	4-Digit 7-Segment Multiplexed Display Shows 4 digits Use code scanning	1-Digit 7-Segment Numeric LED Display Use resistors Check pinout	Dot Matrix Module LED Matrix Display Rows/columns scanned Use driver/code scanning

Inputs / Controls

Red Button Caps For tactile push buttons	Yellow Button Caps For tactile push buttons	Green Button Caps For tactile push buttons	Blue Button Caps For tactile push buttons
Black Button Caps For tactile push buttons	Push Button (Larger)  Normally open Press = connected	Push Button (Smaller)  Normally open Press = connected	Push Button (for caps) Normally open Press = connected
Side Switches  On position = connected Off position = disconnected	IR Receiver Infrared Receiver For remote control Digital output	IR Remote Infrared Remote Control Sends coded IR signals Use with IR receiver	MPR121 Capacitive Touch Controller 12 touch inputs I2C interface
Joystick Module 2-Axis Joystick X/Y analog outputs Push button output	Rotary Encoder Incremental Encoder A/B digital pulses Often includes push button	Keypad Matrix Keypad Row/column scanning Digital inputs	Touch Sensor Capacitive Touch Module Digital output Touch = signal change

Sensors

Ultrasonic Sensor Distance Sensor Trig + Echo pins Measures distance by sound	IR Obstacle Sensor IR Proximity Sensor Adjustable threshold Digital output	Thermistor  Temperature Sensor Resistance changes with temperature	Photoresistor  Light Sensor Resistance changes with light level
---	--	---	---

Water Level Sensor Analog Water Sensor Outputs level voltage Avoid long-term soaking	Temp/Humidity Sensor Environmental Sensor Digital temperature/RH Check pinout	Sound Sensor Module Microphone Sensor Detects sound level Analog/digital output varies	TCS34725 RGB Color Sensor I2C interface Measures color/light
MPU6050 IMU Module Accel + gyro I2C interface	PIR Motion Sensor Infrared Motion Sensor Detects body motion Digital output	Tilt Switch Tilt Sensor Contact switch Detects tilt/vibration	DS3231 RTC Module Real-Time Clock Module I2C Interface Battery Backup Keeps Date And Time
MFRC522 RFID Module 13.56 MHz RFID Reader SPI Interface Use With RFID Tags			

Audio

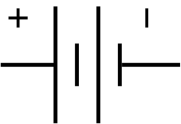
Passive Buzzer Piezo Sounder Needs PWM signal Tone from code	Active Buzzer Self-Oscillating Buzzer No PWM needed DC on = sound	3W 4Ω Speaker Low-Impedance Speaker Use audio amplifier Do not drive from GPIO	2W 8Ω Micro Speaker Low-Impedance Speaker Use audio amplifier Do not drive from GPIO
---	--	---	---

Actuators / Motors

Stepper Motor 4-Phase Stepper Use ULN2003 driver For precise rotation	DC Fan DC Motor Fan Use transistor/driver Do not drive from GPIO	Relay Module Electrically Controlled Switch Isolated load control Digital input	9G Servo PWM Servo Motor Signal + 5V + GND Angle control
--	---	--	---

DC Motor Small Brushed Motor Use Driver / Transistor Do Not Drive From GPIO			
--	--	--	--

Power Accessories

9V Battery  Rechargeable Li-ion Use with 9V clip	Breadboard Power Module Power Supply Module 3.3V / 5V rails Check jumper settings		
--	--	--	--

Connectors / Sockets

40-Pin Headers Breakaway Headers 2.54mm pitch For modules/PCBs	DIP-8 Socket IC Socket For DIP-8 chips Protects ICs	DIP-14 Socket IC Socket For DIP-14 chips Protects ICs	DIP-16 Socket IC Socket For DIP-16 chips Protects ICs
DIP-18 Socket IC Socket For DIP-18 chips Protects ICs	28-Pin ZIF Socket Zero-Insertion-Force Socket For IC testing Lock before use		

Arduino / Prototyping

Arduino Uno R3 Many I/O Pins USB Programmable	Arduino Mega 2560 Many I/O Pins USB Programmable	Prototype Shield Arduino Proto Board For Custom Circuits Fits UNO / Mega Headers	
--	---	---	--

Labels

Resistors / Variable Resistors Fixed + Precision +Adjustable Resistors Includes Trimmer Potentiometers	Capacitors Ceramic + Electrolytic Capacitors Mixed Labeled Values
Inductors Air-Core RF Inductors Measure µH If Needed	Diodes Rectifier / Schottky / Zener Cathode = Stripe Side
BJTs NPN + PNP Transistors Check Pinout Before Wiring	MOSFETs N-Channel + P-Channel MOSFETs Field Effect Transistors
Integrated Circuits Analog, Digital, Driver, Power ICs + Circuit Modules	LEDs / Displays LEDs + Display Modules Use Current-Limit Resistors
Inputs / Controls Buttons, Switches, Keypads User Input Modules	Sensors Environmental / Motion Sensors Check Module Pinouts
Audio Buzzers, Speakers, Audio Modules Do Not Drive Speakers From GPIO	Actuators / Motors Motors, Servo, Relay, Fan Use Drivers When Needed
Power Accessories Batteries, Clips, Adapters Check Voltage And Polarity	Connectors / Sockets Headers, DIP Sockets, ZIF Socket For ICs And Prototyping
Arduino / Prototyping Shields + Arduino Accessories For Custom Circuits	